CS6208 Progress Report

Ask or Act for Cooperative Assistance via Inverse Planning

Team member: Zhu Yufan A0238993J *March 13, 2026*

What Has Been Done

All milestones through March 13 are complete. We implement a cooperative 9×9 two-room gridworld in which a principal holds a hidden goal and issues an ambiguous instruction mapping to $K \in \{1,2,3,4\}$ candidate goals. The assistant maintains a Bayesian posterior over goals, initialised from the instruction and updated by observed principal actions via a Boltzmann likelihood [1, 3], and at each step compares the expected team cost of acting now versus asking the best available clarifying question [2]:

$$\text{CostAsk}(q) = 1+c_q+\mathbb{E}_{a|q}[V(s,b_{q,a})], \quad \text{CostAct} = 1+V(s,b,g^*).$$

Seven policies were implemented: *AskOrAct* (VoI), *NeverAsk*, *AlwaysAsk*, *InfoGainAsk*, *EasyInfoGainAsk*, *RandomAsk*, and *POMCPPlanner*. A full sweep over $\varepsilon \in \{0, 0.05, 0.1\}$, $\beta \in \{1,2,4\}$, 5 replicate seeds $\times$ 20 episodes (100 per cell, 19,800 total) was run with 95% bootstrap CIs. Ablations (question cost, wrong-pick failure mode, deadline margin), robustness sweeps (answer noise, principal β-mismatch), held-out template generalization, and scale-K stress tests (K up to 6) were completed.

Bug discovered and fixed. *AskOrAct* produced results identical to *NeverAsk* at K=2 (zero questions asked). The cause: `ENTROPY_GATE = 0.8`, but the uniform prior over K=2 goals has entropy $\ln 2 \approx 0.693 < 0.8$, so the gate fired at every step. Fix: `ENTROPY_GATE` reduced from 0.8 to 0.3. Post-fix: *AskOrAct* 97% success vs. *NeverAsk* 85% at K=2 (ε=0, β=1).

Preliminary Findings

Figure 1 and Tables 1–2 summarise key results. At K=1 all policies are equivalent (sanity check). At $K \geq 3$, *AskOrAct* outperforms *NeverAsk* with statistical significance: +7.9 pp at K=3 (95% CI $[5.0, 10.8]$) and +8.4 pp at K=4 (95% CI $[5.1, 12.1]$), while asking only 0.6–0.7 questions per episode. *AlwaysAsk* asks 3–4$\times$ more but achieves similar or lower success and much higher regret (+2.1 at K=4). Robustness sweeps show the advantage holds under answer noise (K=4: +4–+9 pp across noise 0–0.2) and principal model mismatch (+13.5 pp when β_{true}=1 vs. $\hat{\beta}$=2). The ablation confirms that a non-zero question cost (c_q=0.5) is essential: setting c_q=0 causes over-asking, reducing success from 73.8% to 69.2%.

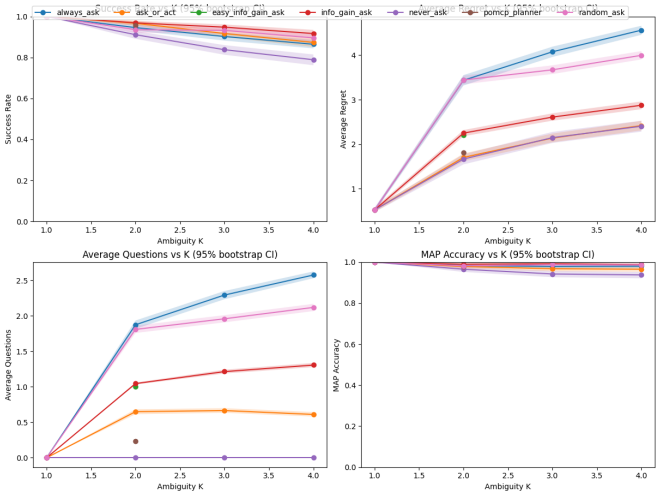

Figure 1: Main evaluation dashboard (100 ep/cell, error bars = 95% CI).

Table 1: Results at K=3, ε=0, β=1 (100 episodes).

Policy	Success	Avg Q	Regret
AskOrAct	**90%**	0.74	**2.19**
NeverAsk	76%	0.00	2.51
InfoGainAsk	93%	1.25	2.58
AlwaysAsk	87%	2.41	4.14

Key Plot/Table for Final Report

Figure 1 (main dashboard) will appear in the final report. The horizontal axis is ambiguity $K \in \{1,2,3,4\}$; the three vertical panels show (top) success rate, (middle) regret, and (bottom) questions asked per episode, one bar group per policy with 95% CI error bars. Readers learn: (a) at K=1 all policies tie—confirming the implementation is correct; (b) the success gap between *AskOrAct* and *NeverAsk* widens monotonically with K; (c) *AlwaysAsk* asks many more questions without a commensurate success gain, visible from the contrast between panels (top) and (bottom). This figure is the central evidence for the claim that selective asking is superior to both extremes.

Aggregate delta table (to appear in final report): rows are $K \in \{2,3,4\}$, columns are ΔSuccess (*AskOrAct*−*NeverAsk*) with 95% CI, ΔRegret (*AskOrAct*−*AlwaysAsk*), and questions/ep. Readers see at a glance whether improvements are statistically significant and at what cost in questions.

Plan for Remaining Time

Step 1 — Failure-mode analysis (*deadline: March 21*). Separate timeout failures from wrong-pick failures per (K, ε, β) for all policies. Identify the specific conditions where *AskOrAct* underperforms (e.g. K=3, noise $= 0.1$ regression) and determine whether they are due to posterior misdirection from noisy answers, exhausted question budget, or a closed ask window. Results will feed directly into the final report's discussion

Table 2: Results at K=2, ε=0, β=1, post-fix (100 ep.).

Policy	Success	Avg Q	Regret
AskOrAct	**97%**	0.71	**1.84**
NeverAsk	85%	0.00	2.16
InfoGainAsk	95%	1.04	2.44
AlwaysAsk	91%	1.97	3.67

section.

Step 2 — Final report (*deadline: April 3*). Write the complete 4-page technical paper: introduction, related work, full method (posterior update equations, VoI decision rule, gating conditions), experiments with aggregate tables and delta-CI analysis, ablation and robustness sections, discussion of limitations, and conclusion. Regenerate all figures from the final corrected CSVs for inclusion.

Step 3 — Presentation (*deadline: April 8*). Prepare 14 slides with actual result numbers, figures, and backup slides for Q&A (entropy gate sensitivity, question cost ablation, scalability).

Step 4 — Submission package (*deadline: April 10*). Run `./run.sh package` to produce `submission_package.zip` containing all data, plots, report PDFs, and reproduction instructions.

AI Tool Declaration

I used Claude Sonnet 4.6 (Anthropic) for coding assistance, debugging, and editing. I am responsible for all research decisions, experimental design, and the content of this report.

References

[1] Chris L. Baker, Rebecca Saxe, and Joshua B. Tenenbaum. Action understanding as inverse planning. *Cognition*, 113 (3):329–349, 2009.

[2] Dylan Hadfield-Menell, Anca D. Dragan, Pieter Abbeel, and Stuart Russell. Cooperative inverse reinforcement learning. In *Advances in Neural Information Processing Systems*, 2016.

[3] Julian Jara-Ettinger, Hyowon Gweon, Laura E. Schulz, and Joshua B. Tenenbaum. The naive utility calculus. *Trends in Cognitive Sciences*, 20(8):589–604, 2016.